

Features

- Output voltage tolerance $\leq \pm 2\%$
- 200mA current capability
- Low-drop voltage
- Very low current consumption: 100 μ A
- Overtemperature protection
- Short-circuit proof
- Suitable for use in automotive electronics
- Green Product (RoHS compliant)

Potential applications

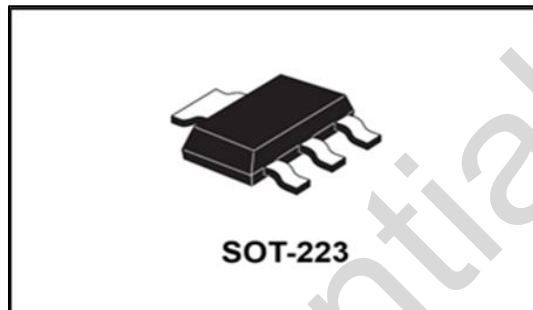
- General automotive applications.

Product validation

- Qualified for automotive applications. Product validation according to AEC-Q100.

Description

The HE4264 is a monolithic integrated low-drop fixed



voltage regulator which can supply loads up to 200 mA. It is functional compatible to the HE4264, but has a reduced quiescent current of typ. 100 μ A. The HE4264 is especially designed for all applications which require very low quiescent currents. The device is available in the small surface mount SOT-223 package. The device is pin compatible to the HE4264. The regulator is designed to supply microprocessor systems under the severe condition of automotive applications and is therefore equipped with additional protection against overload, short-circuit and overtemperature. Of course the HE4264 can be used in all other applications, wherever a stabilized voltage is required.

An input voltage V_{IN} in the range of $5.5\text{ V} < V_{IN} < 40\text{ V}$ is regulated to $V_{OUT, nom} = 5\text{ V}$ with an accuracy of $\pm 2\%$. An accuracy of $\pm 2\%$ is kept for a load current range up to 50 mA.

The device operates in the temperature range of $T_J = -40^\circ\text{C}$ to 150°C .

Part Number	Package	Marking	Packing
HE4264xxxx	SOT-223	HE4264	3000 / Tape & Reel

*Not sure if you got the latest datasheet or not. Please kindly contact the sales of KKChips to confirm. Thanks.

Contents

Contents	2
List of tables	3
List of figures	4
1 Block diagram and pin description	5
1.1 Block diagram	5
2 Pin configuration	6
2.1 Pin assignment.....	6
3 Electrical specifications	7
3.1 Absolute maximum ratings	7
3.2 Functional range	8
3.3 Thermal data	8
4 Electrical characteristics	9
4.1 Electrical characteristics.....	9
4.2 Functional description	10
4.2.1 Application circuit.....	10
5 Typical performance characteristics	11
6 Package information	13
7 Customer Service	15
8 Application information	15
9 Revision history	16

List of tables

Table 2-1 Pin description	6
Table 3-1 Absolute maximum ratings ⁽¹⁾	7
Table 3-2 Functional range	8
Table 3-3 SOT-223 Thermal data	8
Table 4-1 Electrical characteristics voltage regulator	9
Table 6-1 SOT-223 mechanical data	13

List of figures

Figure 1-1 Block diagram	5
Figure 2-1 SOT223 Package	6
Figure 4-1 Application circuit	10
Figure 5-1 Output voltage V_{OUT} versus junction temperature T_j	11
Figure 5-2 Drop voltage V_{dr} versus output current I_{OUT}	11
Figure 5-3 Output voltage V_{OUT} versus input voltage V_{IN}	11
Figure 5-4 Current consumption I_q versus output current I_{OUT}	11
Figure 5-5 Current consumption I_q versus output current I_{OUT}	12
Figure 6-1 SOT-223 package dimensions	13

1 Block diagram and pin description

1.1 Block diagram

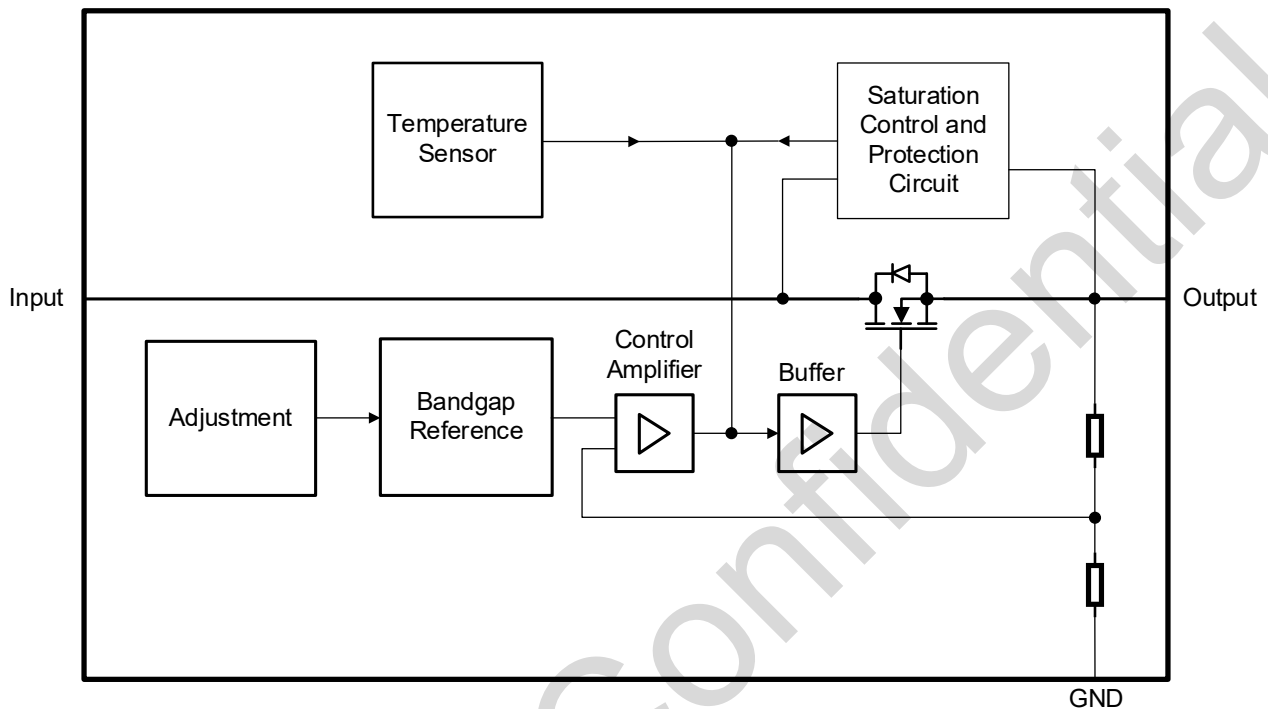


Figure 1-1 Block diagram

2 Pin configuration

2.1 Pin assignment

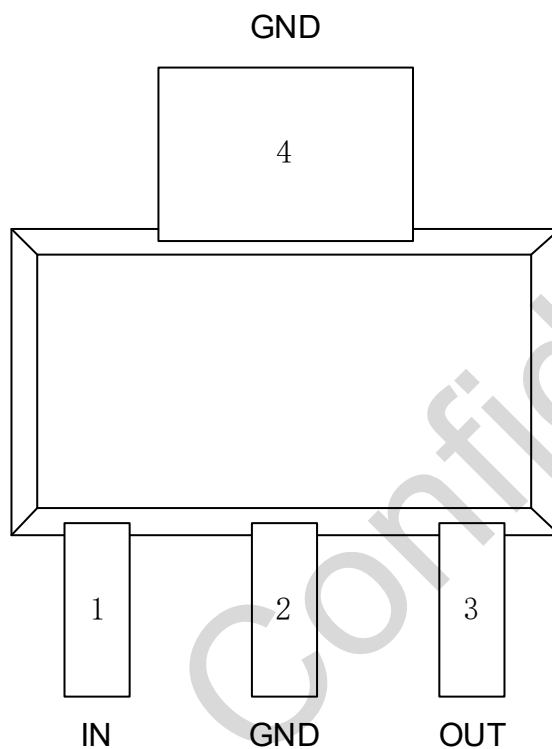


Figure 2-1 SOT223 Package

Table 2-1 Pin description

Pin #	Symbol	Description
1	IN	Input voltage Block to ground directly with a ceramic capacitor
2,4	GND	Ground
3	OUT	5V output voltage Block to ground with a capacitor $CQ \geq 1\mu F$, $ESR \leq 4\Omega$

3 Electrical specifications

3.1 Absolute maximum ratings

$T_j = -40\text{ }^{\circ}\text{C}$ to $150\text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Table 3-1 Absolute maximum ratings⁽¹⁾

Symbol	Parameter	Value		Unit	Number
		Min	Max		
V_{IN}	Input voltage	-0.3	40	V	P_3.1.1
V_{OUT}	Output voltage	-0.3	7	V	P_3.1.2
I_{GND}	Current	50	-	mA	P_3.1.3
T_j	Junction temperature	-	150	$^{\circ}\text{C}$	P_3.1.4
T_{STG}	Storage temperature	-50	150	$^{\circ}\text{C}$	P_3.1.5
$V_{ESD,HBM}$	ESD susceptibility to GND (Human Body Model (HBM)) ⁽²⁾	-2000	2000	V	P_3.1.6
$V_{ESD,CDM}$	ESD susceptibility to GND (Charged Device Model (CDM)) ⁽³⁾	-750	750	V	P_3.1.7

Note:

Stresses above the ones listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods of time may affect device reliability.

Integrated protection functions are designed to prevent IC destruction under fault conditions described in the datasheet. Fault conditions are considered as outside the normal operating range. Protection functions are not designed for continuous repetitive operation.

(1). Not subject to production test, specified by design.

(2). ESD susceptibility, Human Body Model (HBM) according to ANSI/ESDA/JEDEC JS-001 (1.5 k Ω , 100 pF).

(3). ESD susceptibility, Charged Device Model (CDM) according to JEDEC JESD22-C101.

3.2 Functional range

Table 3-2 Functional range

Symbol	Parameter	Value			Unit	Number
		Min	Typ	Max		
V_{IN}	Input voltage range for normal operation	5.5	-	40	V	P_3.2.1
T_j	Junction temperature	-40	-	150	V	P_3.2.2

Note:

Within the functional range, the IC operates as described in the circuit description. The electrical characteristics are specified within the conditions given in the electrical characteristics table.

(1). When V_{IN} is between $V_{IN,ext,min}$ and $V_{OUT,nom} + V_{dr}$, $V_{OUT} = V_{IN} - V_{dr}$. When V_{IN} is below $V_{IN,ext,min}$, V_{OUT} can drop down to 0 V.

(2). The minimum output capacitance requirement is applicable for a worst-case capacitance tolerance of 30%.

(3). Relevant ESR value at $f = 10$ kHz.

(4). Low-temperature enhanced grade product: T_j supports operation down to -55°C (max T_j remains 150°C). This grade requires dedicated component screening and qualification. For orders, please contact kkchips sales team to confirm availability and technical requirements.

3.3 Thermal data

Table 3-3 SOT-223 Thermal data

Symbol	Parameter	Value			Unit	Number
		Min	Typ	Max		
R_{thJA}	Junction to ambient ⁽¹⁾	-	-	164	K/W	P_3.3.1
R_{thJA}	Junction to ambient(300mm ² heat sink area)	-	-	81	K/W	P_3.3.2
R_{thJP}	Junction to Pin ⁽²⁾	-	-	17	K/W	P_3.3.3

(1). Package mounted on PCB 80*80*1.5mm³; 35μ Cu; 5μ Sn; Footprint only; zero airflow.

(2). Measured to pin4.

4 Electrical characteristics

4.1 Electrical characteristics

$V_{IN} = 13.5\text{ V}$; $T_j = -40\text{ }^{\circ}\text{C}$ to $125\text{ }^{\circ}\text{C}$; all voltages with respect to ground, positive current flowing into pin (unless otherwise specified).

Table 4-1 Electrical characteristics voltage regulator

Symbol	Parameter	Condition	Value			Unit	Number
			Min	Typ	Max		
V_{OUT}	Output voltage	$I_{OUT} = 5\text{ mA to }100\text{ mA}$ $V_{IN} = 6\text{ V to }21\text{ V}$	4.85	5.0	5.15	V	P_4.1.1
		$I_{OUT} = 5\text{ mA to }50\text{ mA}$ $V_{IN} = 9\text{ V to }16\text{ V}$	4.9	5.0	5.1	V	P_4.1.2
I_{OUT}	Output-current limiting	-	201	310	420	mA	P_4.1.3
I_q	Current consumption $I_q = I_{IN} - I_{OUT}$	$I_{OUT} = 100\text{ }\mu\text{A}$ $T_j \leq 85^{\circ}\text{C}$ $V_{IN} = 6\text{ V to }18\text{ V}$	-	100	220	μA	P_4.1.4
I_q	Current consumption $I_q = I_{IN} - I_{OUT}$	$I_{OUT} = 100\text{ }\mu\text{A}$ $V_{IN} = 6\text{ V to }18\text{ V}$	-	110	230	μA	P_4.1.5
I_q	Current consumption $I_q = I_{IN} - I_{OUT}$	$I_{OUT} = 50\text{ mA}$ $V_{IN} = 6\text{ V to }18\text{ V}$	-	110	270	μA	P_4.1.6
V_{dr}	Dropout voltage	$I_{OUT} = 100\text{ mA}^{(1)}$	-	0.2	0.5	V	P_4.1.7
$\Delta V_{OUT,IO}$	Load regulation	$I_{OUT} = 1\text{ mA to }100\text{ mA}$ $V_{IN} = 13.5\text{ V}$	-	1	25	mV	P_4.1.8
$\Delta V_{OUT,LI}$	Line regulation	$V_{IN} = 6\text{ V to }28\text{ V}$ $I_{OUT} = 1\text{ mA}$	-	1	25	mV	P_4.1.9
PSRR	Power supply ripple rejection	$f_{ripple} = 100\text{ Hz}$ $V_{ripple} = 0.5\text{ Vpp}$	-	68	-	dB	P_4.1.10
C_{OUT}	Output capacitor	$ESR \leq 4\text{ }\Omega$ at 10 kHz	1	-	-	μF	P_4.1.11

(1) Drop voltage = $V_{IN} - V_{OUT}$ (measured where V_{OUT} has dropped 100 mV from the nominal value obtained at $V_{IN} = 13.5\text{ V}$)

4.2 Functional description

4.2.1 Application circuit

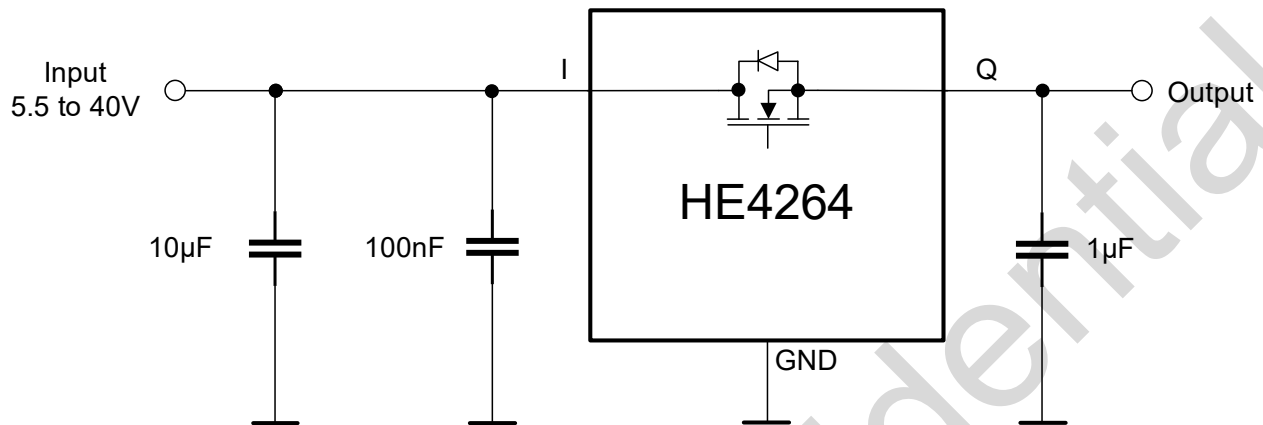


Figure 4-1 Application circuit

The control amplifier compares a reference voltage, which is kept highly precise by resistance adjustment, to a voltage that is proportional to the output voltage and drives the gate of the MOS via a buffer. Saturation control, working as a function of load current, prevents any over-saturation of the power element.

The IC is additionally protected against overload and overtemperature.

In the HE4264 the output voltage is divided and compared to an internal reference of 1.2 V typical. The regulation loop controls the output to achieve an output voltage of 5 V with an accuracy of $\pm 2\%$ at an input voltage range of $5.5 \text{ V} < V_{\text{IN}} < 40 \text{ V}$.

Figure 4-1 shows a typical application circuit. For stability of the control loop the HE4264 output requires an output capacitor C_{OUT} of at least $1\mu\text{F}$ with a maximum permissible ESR of 4Ω . Tantalum as well as multilayer ceramic capacitors are suitable.

At the input of the regulator an input capacitor is necessary for compensating line influences (100 nF ceramic capacitor recommended). A resistor of approx. 1Ω in series with C_{IN} , can damp any oscillation occurring due to the input inductivity and the input capacitor.

In the application circuit shown in Figure 4-1 an additional electrolytic input capacitor of $470 \mu\text{F}$ is added in order to buffer supply line influences. This capacitor is recommended, if the device is sourced via long supply lines of several meters.

The HE4264 can supply up to 200 mA. However for protection for high input voltage above 21 V, the output current is reduced (SOA protection).

5 Typical performance characteristics

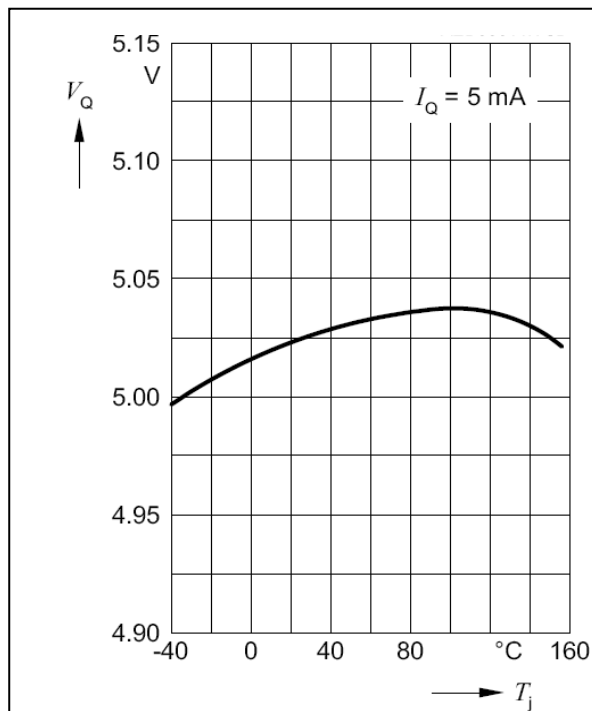


Figure 5-1 Output voltage V_{OUT} versus junction temperature T_j

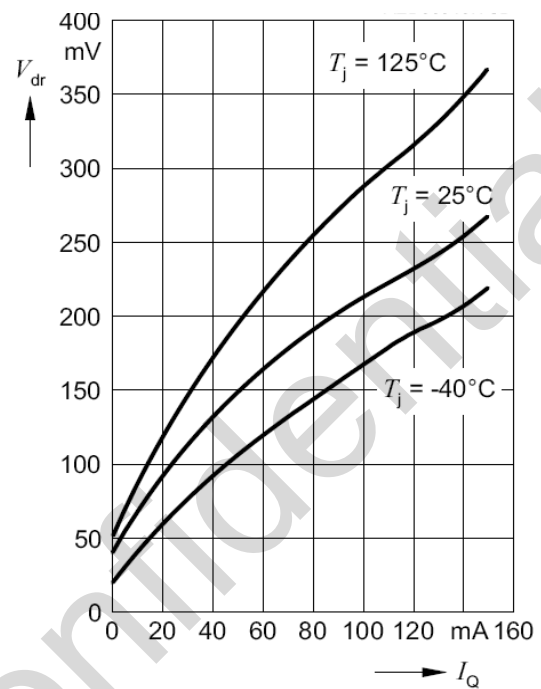


Figure 5-2 Drop voltage V_{dr} versus output current I_{OUT}

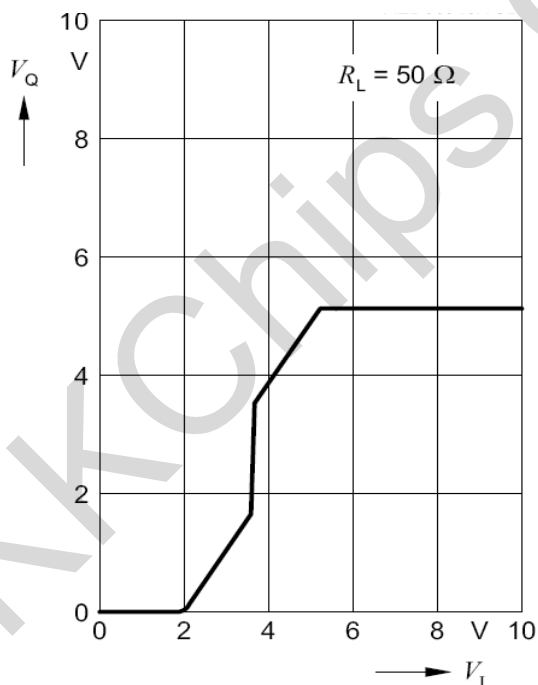


Figure 5-3 Output voltage V_{OUT} versus input voltage V_{IN}

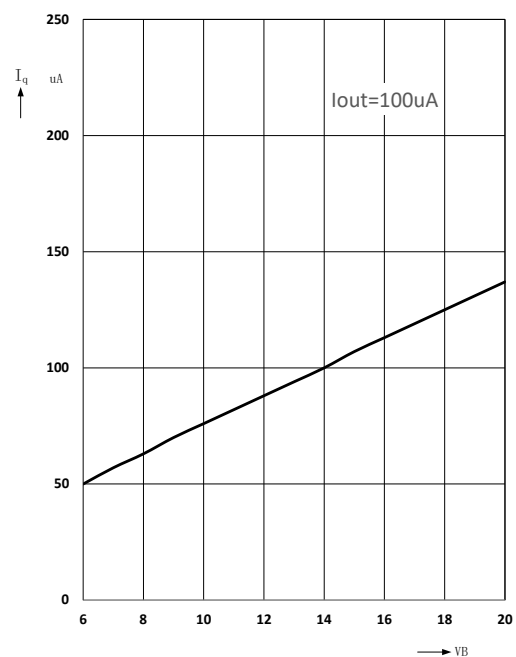


Figure 5-4 Current consumption I_q versus output current I_{OUT}

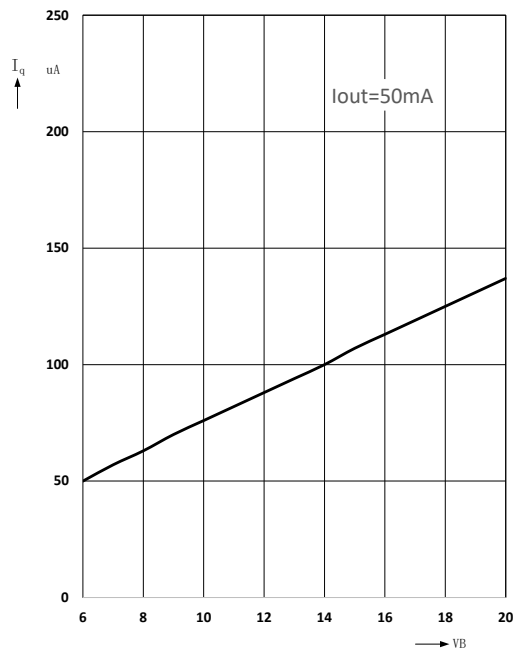


Figure 5-5 Current consumption I_q versus output current I_{out}

6 Package information

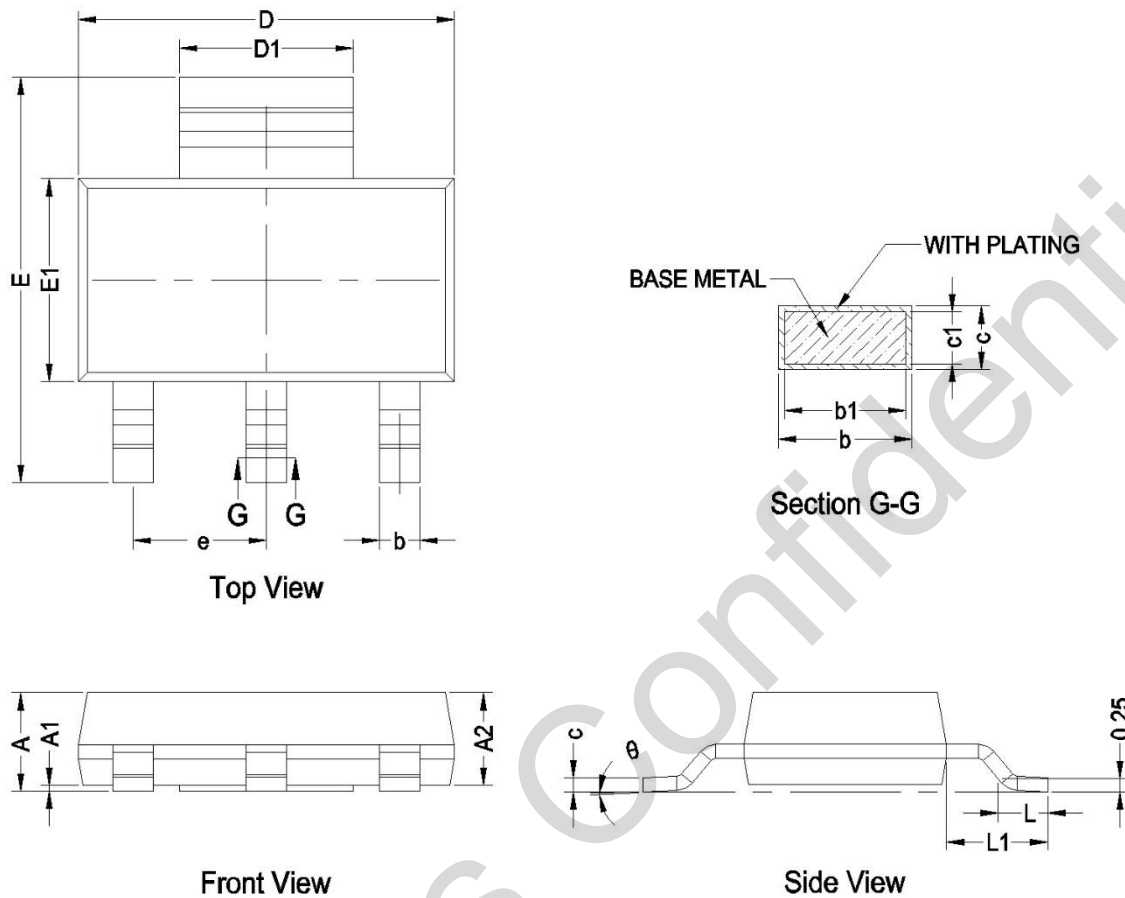


Figure 6-1 SOT-223 package dimensions

Table 6-1 SOT-223 mechanical data

Dim.	mm		
	Min.	Typ.	Max.
A	1.50	1.65	1.80
A1	0.03	0.06	0.09
A2	1.50	1.60	1.70
D1	2.90	3.00	3.10
b	0.69	-	0.77
b1	0.68	0.71	0.74
c	0.30	-	0.34

HE4264

Low dropout linear voltage regulator



Electrical characteristics

c1	0.29	0.30	0.31
D	6.40	6.50	6.60
E	6.80	7.00	7.20
E1	3.40	3.50	3.60
e	2.30 BSC		
L	0.90	-	1.15
L1	1.75REF		
θ	0°	-	7°

7 Customer Service

Guangdong KKChips Automotive Electronics Tech Co., Ltd.

9th Floor, Grandtek Building B2, No. 18 Kexue Avenue, Huangpu District, Guangzhou

8 Application information

The following information is given as an example for the implementation of the device only and shall not be regarded as a description or warranty of a certain functionality, condition or quality of the device.

KKChips Confidential

9 Revision history

Date	Revision	Changes
2026/05/25	0.0.1	Initial release.